The assigning rule is

$X(1) = 1$, $X(2) = 2$, $X(3) = 3$, $X(4)=4$, $X(5)=5$ and $X(6)=6$

Hence the values taken by the random variable X are 1,2,3,4,5,6. These values are also called the realization of the random variable X.

(ii) Random experiment : Two coins are tossed simultaneously.

Sample space : $S=\{HH, HT, TH, TT\}$

Assigning rule : Let X be a random variable defined as the number of heads comes up.

Sample Point ω	HH	HT	TH	TT
X (ω)	2	1	1	0

Here, the random variable X takes the values 0, 1, 2.

(iii) Experiment : Two dice are rolled simultaneously.

Sample space : $\{(1, 1),(1, 2),(1, 3),\ldots(6, 6)\}$

Assigning rule : Let X denote the sum of the numbers on the faces of dice

then $X_{ij} = i + j$, Here, i denotes face number on the first die and j denotes the face number on the second die.

Then X is a random variable which takes the values 2, 3, 4 . …. 12.

That is the range of X is {2, 3, 4…… 12}

9.2 Discrete and Continuous random variables

Random variables are generally classified into two types, based on the values they take such as Discrete random variable and Continuous random variable.

9.2.1 Discrete random variable

A random variable is said to be discrete if it takes only a finite or countable infinite number of values.